# ST512-SP26-Homework-1

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### Problem 1

n = 36

$\bar{x}$ = 23.4

sd (s) = 7.2

CI: $\bar{x}\pm$ t* x SE

SE = $\frac{s}{\sqrt{n}}$

SE = $\frac{7.2}{\sqrt{36}}$ = 1.2

t* = qt(1-(.04/2), 36) = 2.13

CI: 23.4 $\pm$ 2.13 * 1.2

96% CI: (20.84,25.96)

### Problem 2

$H_o : \mu = 3.5$

$H_A : \mu < 3.5$

Will be conducting a left-tailed test because we want to know if the drug delivers relief faster (i.e., less than 3.5 minutes).

n = 50

$\bar{x}$ = 3.1

s = 1.5

$\alpha$ = 0.05

SE = $\frac{s}{\sqrt{n}} = \frac{1.5}{\sqrt{50}}$ = 0.212

t-stat = $\frac{\bar{x}-\mu_o}{SE}$ = $\frac{-0.4}{0.212}$ = -1.887

p-value = pt(t.value, df) = 0.033

p-value < $\alpha$, we reject $H_o$

Therefore, there is sufficient evidence at the 5% confidence interval to conclude that the newly developed pain reliever delivers perceptible pain relief more quickly than the standard pain reliever.

## Problem 3

| | Design 1 | Design 2 |
|---|---|---|
| $n_i$ | 11 | 6 |
| $\bar{y}_i.$ | 52 | 46 |
| sd | 12 | 10 |

### a

point estimate: $\mu_1 \hat{-} \mu_2$ = $\bar{y}_1$. - $\bar{y}_2$. = 52 - 46 = 6

95% CI: ($\bar{y}_1$. - $\bar{y}_2$.) $\pm$ t* x SE

Assuming equal variance:

df = $n_1 + n_2$ - 2 = 15

$s_p^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{df} = \frac{10(12^2)+5(10^2)}{15}$ = 129.33

$s_p = \sqrt{129.33}$ = 11.37

SE = $s_p\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}$ = 11.38$\sqrt{\frac{1}{11}+\frac{1}{6}}$ = 5.78

t* = qt(0.975,15) = 2.13

95% CI: 6 $\pm$ 2.13 * 5.78

95% CI: (-6.31, 18.31)

### b

$H_o : \mu_1 = \mu_2$

$H_A : \mu_1 \neq \mu_2$

t-stat = $\frac{pointestimate-0}{SE}$ = $\frac{6}{5.78}$ = 1.04

p-value: 2 * p(T > |t-value|) = 0.31

$\alpha = 0.01$

p-value $> \alpha$; fail to reject $H_o$

There is not sufficient evidence at the 1% significance level to conclude the mean monthly sale differ between the two designs.

## R Code

```
qt(1-(.04/2), 36) #problem 1
```

```
[1] 2.130871
```

```
pt(-1.887,49) #problem 2
```

```
[1] 0.03254595
```

```
qt(0.975,15) #problem 3.a
```

```
[1] 2.13145
```

```
2*(1-pt(1.04,15)) #problem 3.b
```

```
[1] 0.314814
```